

Observation of continuous time crystals and quasi-crystals in spin gases

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Continuous time crystals (CTCs) are a phase of matter characterized by spontaneous breaking of continuous time-translation symmetry. Recently, CTCs have garnered interest due to breakthroughs in experimental implementation. Here we report the experimental observation of CTCs in noble-gas nuclear spins and uncover previously unexplored dynamical phenomena. We observe that the CTCs manifest as persistent limit cycle oscillations of nuclear spins, with coherence times exceeding hours. Notably, these oscillations are robust against noise perturbations and exhibit random time phases upon repetitive realization, epitomizing continuous time-translation symmetry-breaking intrinsic to CTCs. Additionally, we observe a dynamical phase featuring quasi-periodic oscillations and random time phases, indicating the emergence of the continuous time quasi-crystals proposed by recent theories. By varying the feedback strength and magnetic gradient, we observe complex dynamical phase transitions between time crystal phases and chaotic regimes. This work broadens the catalog of phases of spin gases and unlocks opportunities in precision measurements.

Time crystals, which exhibit broken time-translation symmetry analogous to the broken space-translation symmetry in ordinary crystals, have captured interest from the physics community since their initial proposal. The original concept of time crystals involved closed systems and demonstrated continuous time-translation symmetry breaking through oscillatory dynamics^{1,2}. However, a no-go theorem soon revealed that such time crystals are prohibited by nature^{3–5}, prompting scientists to explore new possibilities. It has been theoretically argued that long-range interacting models in closed quantum systems could circumvent the no-go theorem⁶, and this model requires further exploration. One promising direction extends the concept of time crystals into periodically driven systems, known as discrete time crystals (DTCs)^{7–11}. They break the discrete time-translation symmetry imposed by periodic external driving and are characterized by sub-harmonic oscillations. Numerous experimental realizations of DTCs

have emerged^{12–21}. Recently, another approach using open systems to circumvent the no-go theorem has gained attention^{22–25}. This approach allows for continuous time crystals (CTCs) that continuously break time-translation symmetry, closely aligning with the spirit of the original proposal. To date, CTCs have been realized on several platforms, including atomic Bose-Einstein condensate²⁶, metamaterial nanostructure²⁷, semiconductor²⁸, polariton condensate²⁹, and Rydberg gas³⁰. Currently, exploring time crystals across a diverse range of physical systems and uncovering their complex properties is at the forefront of scientific research.

Here we report the experimental observation of a CTC in noble-gas nuclear spins. Our experiments use ¹²⁹Xe noble gas as a testbed, which interacts with overlapping ⁸⁷Rb gas. The ⁸⁷Rb gas measures the ¹²⁹Xe spin signal and feeds this signal back to ¹²⁹Xe through feedback coils. This feedback mechanism induces nonlinear interactions

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between ^{129}Xe atoms, which are essential for the generation of CTCs. In such a noble gas, we observe limit cycle oscillations as signatures of a CTC and demonstrate their spontaneous breaking of continuous time-translation symmetry through random time phase distribution in repetitive measurements²⁶. Interestingly, while magnetic field inhomogeneities are typically seen as detrimental due to their dephasing effects on spin relaxation, we introduce a gradient magnetic field in experiments, which enriches the dynamics of the noble gas. In this situation, we observe a dynamical phase featured by quasi-periodic oscillations and random time phases, indicating the emergence of the continuous time quasi-crystals (CTQCs). We investigate phase transitions of the time crystal across system parameters such as feedback strength and magnetic gradient. The phase diagram shows that the time crystal phase spans a wide range of parameters. Additionally, our time crystal can persist its oscillation pattern against temporal perturbations, verifying the robustness. Our work paves paths for realizing and exploring time crystals, offering opportunities for applications in fields such as quantum metrology.

Results

Our system, exemplified by the Rb-Xe configuration, integrates overlapping noble gas and alkali-metal gas. In the setup depicted in Fig. 1, the ^{129}Xe nuclear spins are continuously optically pumped through spin exchange with embedded ^{87}Rb ^{31,32}. These ^{87}Rb atoms are polarized by a circularly polarized laser beam tuned to the Rb D1 transition at 795 nm. A linear gradient magnetic field is applied along the same direction as the Rb pump beam (z-axis), resulting in a continuous Larmor frequency distribution $\rho(\omega)$. The ^{129}Xe nuclear spin component is measured via interactions with the ^{87}Rb spins using a linearly polarized probe beam oriented along the x-axis³³ (see Supplementary Fig. 1). The intensity change of the probe beam, which is proportional to the average ^{129}Xe spin polarization $\bar{P}_x = \int \rho(\omega) P_x(\omega) d\omega$, is recorded by a photodiode. The corresponding ^{129}Xe signal is fed back to the nuclear spins as a magnetic field along y, expressed as $B_f = -\alpha \bar{P}_x / \gamma$ where α represents the feedback strength and γ is the gyromagnetic ratio of ^{129}Xe . More details about our feedback scheme can be found in the Supplementary Information.

When the feedback strength α is weak, the noble-gas nuclear spins exhibit a normal phase, with the spins polarized along the bias field. When the nonlinear interaction is sufficiently strong, three distinct dynamical phases emerge: limit cycle, quasi-periodic, and chaos. The limit cycle phase, which corresponds to the synchronization across different frequencies, is characterized by long-lived periodic oscillations, each with a singular, steady frequency. The quasi-periodic oscillations are superimposed by oscillators of two incommensurate frequencies ω_s and ω_τ , where the value of ω_τ depends on the feedback strength and gradient field³⁴. When the ratio of ω_s/ω_τ is irrational, these spin oscillations are ordered but not apparently periodic, different from the limit cycle phase. The application of a gradient magnetic field is crucial for the emergence of this spin phase. To illustrate the different dynamical behaviors of these phases, three representative experimental examples are presented below. In order to gain deeper insights into the spin dynamics, we numerically investigate the stable dynamics of the three corresponding phases.

We numerically simulate the phase portraits of the average spin polarization $\{\bar{P}_x, \bar{P}_y, \bar{P}_z\}$ in three phases using the equations described in the Methods section. Additionally, Poincaré sections, defined by the intersections at $P_y = 0$, are accentuated in the phase portrait figures to offer a more clear perspective. The phase portrait of the limit cycle is a closed loop, and its Poincaré section consists of two discrete points (Fig. 2a). They both represent a periodic oscillation in which the system revisits the same state after each period. The phase portrait of quasi-periodic phase traces a torus that never closes, implying the absence of a true period (Fig. 2b). Correspondingly, the Poincaré section forms closed curves, corroborating the quasi-periodic nature of the system

by illustrating the regular yet non-repeating interplay of incommensurate frequencies. In the chaotic phase, the phase portrait reveals irregular surfaces formed by a dense, scattered distribution of points (Fig. 2c). The dense distribution of points reflects the underlying chaos, where minor perturbations in initial conditions induce substantial trajectory divergence. Additionally, the corresponding Poincaré section exhibits a self-similar pattern of densely scattered points, providing further evidence of the chaotic behavior. Next, we present the experimentally measured signal of the one-dimensional component $\bar{P}_x$.

In the experiment, after a transient build-up process, the system levels into stable oscillations and maintains stable dynamic behaviors (see Supplementary Fig. 4). For demonstration, we extract 1000-s segments from the steady states of the three distinct dynamical phases. Figure 2d shows a representative time trace of the probe intensity signal in the stationary window, along with the corresponding spectrum for the limit cycle phase. The time trace reveals impeccably periodic oscillations, showcasing both a stable amplitude and frequency. By performing a fast Fourier transformation on the data recorded during this interval, we unveil its Fourier spectrum, which is dominated by a single, narrow peak. The linewidth of this peak is constrained solely by the duration of the acquisition time, underscoring the precision and stability of the oscillatory behavior. The observed limit cycle phase emerges through spontaneous breaking of continuous time-translation symmetry (as discussed below), and is a distinctive signature of the CTC.

Figure 2e illustrates a quintessential example of the quasi-periodic phase observed in our experiment. In this phase, the Fourier spectrum reveals peaks that are equidistantly spaced. The time traces are ordered but non-periodic, which are characteristic of recently proposed time quasi-crystal^{35–37}. As an example, ref. 35 revealed the emergence of a time quasi-crystal in the presence of a “staircaselike” on-site potential. It is worth noting that a similar quasi-periodic oscillation has recently been observed in an atom-cavity system³⁸. In contrast to the recently demonstrated discrete time quasi-crystals^{39,40}, our observation of the quasi-periodic phase suggests the emergence of CTQCs.

A typical example of a chaotic phase is shown in Fig. 2f. The temporal signal is composed of wave packets, but the period and amplitude of these wave packets vary randomly over time. The corresponding Fourier spectrum features broadened peaks and noisy wings. To confirm the chaotic nature of this phase, we employ the Chaos Decision Tree Algorithm (see Supplementary Information)⁴¹. The chaos criteria processes time series recordings and outputs a single value K , which approaches 1, signifying chaotic systems, whereas a value nearing 0 indicates periodic or quasi-periodic systems. The value of K is approximately 0.9819 for the chaotic phase shown as Fig. 2f. For comparison, the value of K is approximately 0.0016 for the limit cycle phase as depicted in Fig. 2d, and 0.0028 for the quasi-periodic phase as depicted in Fig. 2e. For our data, this method is consistent with the maximum Lyapunov exponent and correlation dimension in identifying chaotic behavior (see Supplementary Figs. 6 and 7). The observed chaotic oscillations disrupt the ideal periodicity or quasi-periodicity over time and can be interpreted as the “melting” of the time-ordered phases, including the CTC and CTQC.

We investigate the spontaneous breaking of continuous time-translation symmetry in CTCs and CTQCs. To confirm that this symmetry breaking occurs spontaneously, the periodic oscillation of a noble gas should manifest at any arbitrary initial phase, leading to a random relative time phase across various repeated realizations^{26,27}. To verify that the oscillations in our experimental system exhibit arbitrary time phases, we repeatedly measure the limit cycle and quasi-periodic phases under fixed experimental parameters. For comparison, the time phase in the chaotic phase is also repeatedly measured. The measured time phase of the limit cycle is shown in Fig. 2g, displaying a

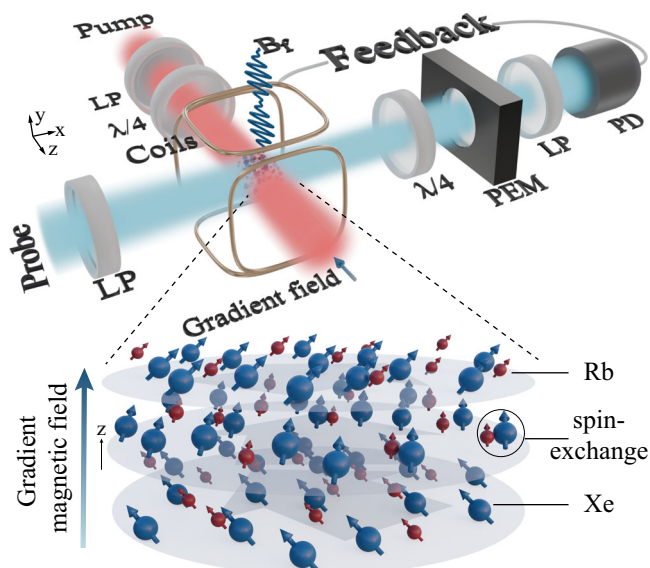


Fig. 1 | Schematic of the experimental setup. The ^{129}Xe atoms are polarized and detected through spin exchange collisions with optically pumped ^{87}Rb , where ^{87}Rb atoms are optically polarized by a circularly polarized pump beam and detected via a rotation of the polarization of a linearly polarized probe beam. The detected information of ^{129}Xe spins is magnetically coupled to a feedback circuit, which feeds back real-time B_y through coils along y and induces nonlinear dynamics in ^{129}Xe spins. The coils along z generate an inhomogeneous bias field B_z , resulting in a continuous Larmor frequency distribution. LP linear polarizer, $\lambda/4$ quarter-wave plate, PD photodiode, PEM photoelastic modulator.

random distribution between $[0, 2\pi]$, corresponding to complex Fourier amplitudes around a loop. This confirms the spontaneous breaking of continuous time-translation symmetry in the limit cycle phase. Accurate feedback activation is essential for determining the relative time phase. To achieve this, we design a high-precision feedback switch control circuit along with synchronized data acquisition (see Supplementary Information). The quasi-periodic phase exhibits similar properties, with the complex Fourier amplitude at each frequency randomly distributed on the respective loop, and their time phases covering $[0, 2\pi]$ (Fig. 2h). Our numerical simulations show that the 2π random phase distribution could arise from initial fluctuations (see Supplementary Fig. 2). This distribution emerges due to the spontaneous breaking of continuous time-translation symmetry, and not all self-oscillations in nonlinear systems exhibit this characteristic 2π random phase behavior (see Supplementary Fig. 3).

In contrast, the chaotic phase displays distinctly different characteristics, lacking stable oscillation. In a chaotic noble-gas system, a Fourier peak may not consistently appear in other intervals of a single realization or across different realizations. As a result, when attempting to calculate the phase of the Fourier amplitude at the same frequency in repeated measurements of chaotic systems, it is often impossible to obtain the same peak's phase for each measurement. The distribution of the calculated time phase in chaos is shown in Fig. 2i, where all points scatter across a circular surface. This scattering occurs because most points do not correspond to a peak but rather to the broadened baseline or noisy wings.

We now investigate the phase transition as feedback strength increases. Figure 3a presents the phase diagram derived from the Fourier spectra at various feedback strengths α , with a fixed magnetic gradient $g \approx 50$ nT/cm. Initially, under weak feedback, the phase of the noble gas remains stable at a no-signal fixed point. Upon surpassing a critical feedback strength (in our experiment, this threshold is $\alpha \approx 9.57$ rad/s), the limit cycle phase emerges, although the oscillation amplitude diminishes as the feedback increases. As the feedback

strength continues to rise, the quasi-periodic phase emerges, revealing an increasing number of harmonics. For example, at $\alpha \approx 265.77$ rad/s, around 17 peaks can be distinctly identified, each possessing nearly identical linewidths, below 1 mHz. The frequency comb gradually stretches with increasing feedback. With even stronger feedback, chaotic oscillations manifest, rendering the spectrum blurred. It is crucial to note that these observations are made in a gradient field; without this gradient field, neither quasi-periodic nor chaotic phases are observed.

We observe an unusual phase transition from a chaotic phase back to CTC phases. Previous works only reported the melting of the time crystal into a chaotic phase without noting this reverse transition. In contrast, our investigation under various magnetic gradients uncovers this anomalous behavior. As depicted in Fig. 3b, the CTC phase undergoes a cycle of formation, “melting”, and re-formation with increasing magnetic gradient g . Initially, similar to the phase transition pattern observed with varying feedback strength, the periodic oscillation of the noble-gas spins emerges. As the magnetic gradient g increases, the oscillation amplitude slightly diminishes and the range extends. Then, chaos abruptly emerges at $g \approx 80$ nT/cm, indicated by a sudden change in the K value. However, in a distinct deviation from previous observations, the chaotic phase disappears, followed by the re-emergence of a limit cycle phase above a critical value of $g \approx 92$ nT/cm. These dynamical transitions can be understood as bifurcations or as the onset and breakdown of synchronization. For example, at large magnetic field gradients, the broader frequency distribution $\rho(\omega)$ decreases the number of spins contributing to each frequency mode, thereby facilitating resynchronization and enabling the re-emergence of a limit cycle from chaos. Similar transitions have been observed in other nonlinear systems—for instance, in the Lorenz system, increasing the Rayleigh parameter r beyond a critical threshold induces a transition from chaotic behavior back to a limit cycle^{42,43}.

In order to more comprehensively demonstrate the dependence of the time crystal behavior on the feedback strength α and the field gradient g , a phase diagram spanning the two-dimensional parameter space (α, g) is provided in the Supplementary Fig. 5. It is worth noting that, while recent theoretical work (ref. 34) explores similar phase transitions between limit cycles, quasi-periodicity, and chaos, it based on simple models that differ from our experimental conditions, leading to qualitatively different outcomes.

Having demonstrated the spontaneous breaking of continuous time-translation symmetry, we further verify the time crystalline nature of our system by examining its robustness against noise—a distinctive and desirable feature of time crystals that stems from the feedback-induced nonlinearity. We demonstrate this robustness by showing the persistence of the oscillation pattern against temporal perturbations. In experiments, we introduce white noise of varying intensities into the transverse feedback field as a test. Figure 4 illustrates the intensity spectrum at different levels of magnetic noise. In the low noise regime, the intensity signal retains the fundamental oscillation pattern similar to the noise-free situation. As the noise strength increases, additional noisy peaks emerge, yet the fundamental oscillation pattern remains clearly recognizable. With further increases in noise strength, the noise gradually begins to dominate the ordered oscillation. The corresponding K value (see inset) shows how increasing noise strength “melts” the time crystal. These observations demonstrate that the observed time crystal can withstand a range of noise levels. This, combined with the persistence of the time crystal across a wide range of system parameters, confirms the robustness of the observed time crystal.

Discussions

Time crystals can manifest within nonlinear dynamical systems. However, not all nonlinear auto-oscillation behaviors qualify as CTCs. While

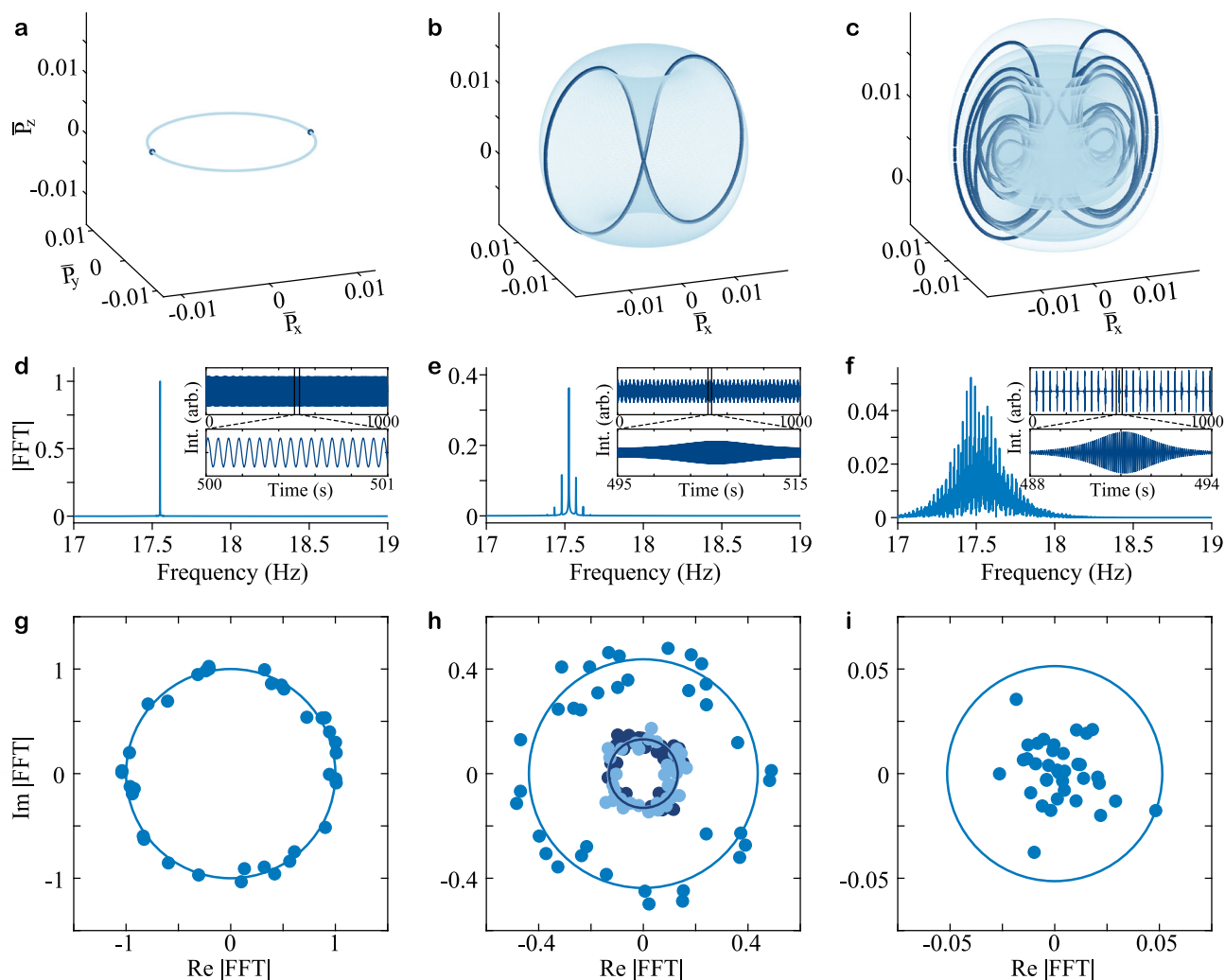


Fig. 2 | Representative dynamical behaviors. **a–c** Phase portraits of the average spin polarization $\{P_x, P_y, P_z\}$ and corresponding Poincaré sections (accentuated) at $P_y = 0$ from numerical simulation. **d–f** Fourier spectrum (normalized) obtained through the fast Fourier transform (FFT) of the intensity signal recorded by PD in the stationary window (inset). **g–i** Relative time phase distribution. Each point

represents the Fourier amplitude at the same peak frequency on the complex plane. Each experiment is performed at $\alpha \approx 42.52$ rad/s, $g \approx 50$ nT/cm for the Limit cycle phase; $\alpha \approx 159.62$ rad/s, $g \approx 50$ nT/cm for the Quasi-periodic phase; $\alpha \approx 350.86$ rad/s, $g \approx 50$ nT/cm for the Chaotic phase.

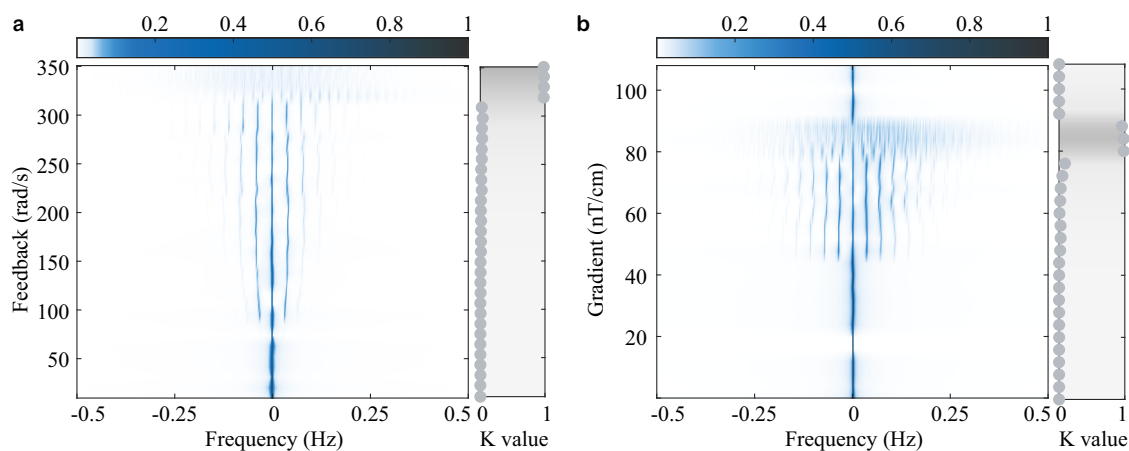


Fig. 3 | Experimental phase diagram and transition. **a** The phase diagram and K dependence on feedback strength α at a fixed magnetic gradient $g \approx 50$ nT/cm. The frequency in the phase diagram represents the shift relative to the central

frequency. **b** The phase diagram and K dependence on magnetic gradient g at a fixed feedback strength $\alpha \approx 276.48$ rad/s.

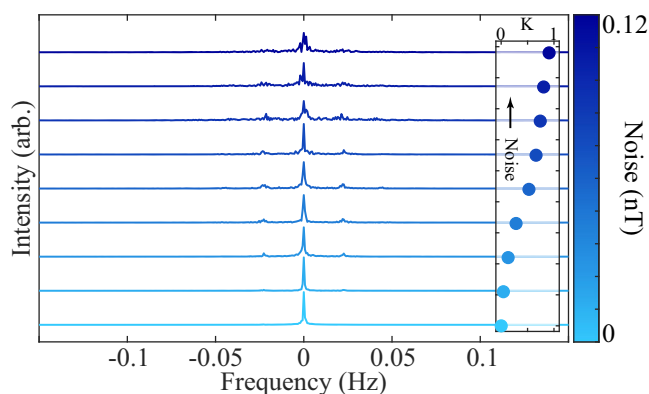


Fig. 4 | Robustness against temporal perturbations. Fourier spectrum and corresponding K value (inset) of the probe beam intensity signal at increasing noise strength, performed at fixed $\alpha = 42.52$ rad/s, $g = 50$ nT/cm. The light blue spectrum at the bottom is the noise-free case, and the deepening of blue corresponds to the increase in the noise strength. The frequency represents the shift relative to the central frequency.

CTCs share certain properties with general nonlinear systems, such as robustness against noise, it is the spontaneous symmetry breaking of continuous time-translation symmetry that fundamentally distinguishes them. In this work, the criterion for CTCs associated with spontaneous breaking of continuous time-translation symmetry is defined following ref. 26, and is manifested as the random and uniform distribution of oscillation phases. Distinctively, in the DTC scenario, the phase distribution can become non-uniform or concentrated in particular sectors of the circle, as described in ref. 44. The rich spectrum of dynamical phases and the transitions observed in our system naturally arise from bifurcations within the nonlinear dynamics framework. Alternatively, these phenomena may be further interpreted with the synchronization theory, which offers a unifying perspective on complex collective behavior in coupled oscillator systems^{45,46}. In our setup, the applied magnetic field gradient introduces a continuous distribution of Larmor frequencies ω , effectively treating each nuclear spin mode as an individual oscillator. These modes are globally coupled via the nonlinear feedback field (with strength α), while the gradient (characterized by g) imposes a frequency detuning across the ensemble—analogue to the mismatch of intrinsic frequencies in canonical synchronization models. The interplay between nonlinear coupling and detuning governs the emergence and breakdown of synchronization, giving rise to the observed dynamical phases. The limit cycle phase corresponds to a synchronized state with a dominant frequency, whereas quasi-periodic and chaotic dynamics emerge when synchronization is frustrated.

Our work expands the range of physical platforms for realizing time crystals and enriches the exploration of their properties. We introduce a key ingredient—a magnetic field gradient—that enriches the dynamical behavior of the noble-gas spin system, enabling the emergence of quasi-periodic and chaotic phenomena. While our demonstration focuses on an Rb-Xe hybrid gas system, we believe that integrating magnetic field gradients with feedback, as demonstrated in our work, is widely applicable across various platforms. This includes systems such as alkali-metal atomic gases, nuclear spin liquids⁴⁷, and others. Furthermore, although we adopted the current feedback technique^{48,49}, we believe our methodology is equally extensible to other feedback mechanisms. This implementation of time crystals would open up possibilities for exploring non-equilibrium phases of matter. Notably, due to experimental constraints such as limited lifetimes, chaos has rarely been observed in CTC research^{26,27,29,30}, with a recent example documented only in an electron-nuclear spin system in semiconductors²⁸.

Beyond its intrinsic conceptual interest, our work on time crystals holds practical value in quantum metrology. For example, the limit cycle phase and the corresponding precise control of feedback in our CTCs represent a crucial class of spin masers^{48,49}. Spin masers, with their long-lived persistent oscillations and robustness, can be widely applied in various fields, such as high-precision magnetometry⁴⁹, searches for electric dipole moments⁵⁰, tests of fundamental symmetry⁵¹, and probes of new physics⁵². Our spin maser is particularly notable for oscillating at low frequencies (approximately 10 Hz) while maintaining high relative frequency accuracy. This makes it ideal for experimental investigations on fundamental questions in physics, where associated frequencies or frequency shifts are small⁵¹. Furthermore, existing masers lack diverse dynamic phases, which limits further exploration. In contrast, the quasi-periodic phase with split frequencies in our CTQCs exhibits richer dynamic behavior, successfully enabling the creation of a multimode maser.

Methods

Under the feedback scheme, the system dynamics are governed by a Hamiltonian that includes Zeeman interactions with the gradient and feedback magnetic field, spin exchange interactions with polarized ⁸⁷Rb atoms, and relaxation mechanisms. Further details can be found in the Supplementary Information. Neglecting quantum fluctuations, the dynamics of ¹²⁹Xe spins can be described by the evolution of spin operator expectation values. Here, the spin polarization is defined as the normalized expectation value of ¹²⁹Xe spin operator, $\mathbf{P} = 2\langle\mathbf{K}\rangle$. For our spin-1/2 system, the evolution of the polarization can be formally expressed as a set of nonlinear Bloch equations:

$$\begin{cases} \dot{P}_x(\omega) = \omega P_y(\omega) + \alpha P_z(\omega) \int \rho(\omega') P_x(\omega') d\omega' - \frac{P_x(\omega)}{T_2} \\ \dot{P}_y(\omega) = -\omega P_x(\omega) - \frac{P_y(\omega)}{T_2} \\ \dot{P}_z(\omega) = \alpha P_x(\omega) \int \rho(\omega') P_x(\omega') d\omega' - \frac{P_z(\omega)}{T_1} + R_{SE} \end{cases} \quad (1)$$

where $T_1(T_2)$ is ¹²⁹Xe longitudinal (transverse) relaxation time, R_{SE} is the pumping rate through spin exchange collisions with ⁸⁷Rb. The feedback-induced nonlinear terms, $\alpha P_z(\omega) \int \rho(\omega') P_x(\omega') d\omega'$ and $\alpha P_x(\omega) \int \rho(\omega') P_x(\omega') d\omega'$, can give rise to auto-oscillations that may exhibit collective synchronization, multi-frequency dynamics, or chaotic behavior. These phenomena can be understood within the general framework of the dynamics of nonlinear oscillating systems^{42,43,45,46}. Using this set of equations, we investigate alternative feedback configurations, for instance by applying feedback along different magnetic field directions, in which the dynamical phases reported in this work are no longer observed (see Supplementary Fig. 8).

Data availability

The data generated in this study are provided in the Source data file. Source data are provided with this paper.

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Author contributions

Y.H. and T.W. designed and performed experiments, analyzed the data, and wrote the manuscript; H.Y. analyzed the data; M.J., Z.L., and X.P.

proposed the experimental concept, devised the experimental protocols, and wrote the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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